

5. Recursive Processing Analysis

Observation-based full translation

ONE-LINE SUMMARY Within the scope observed here, arrangement information inside inherited materials does not appear to be uniformly expanded recursively. Reference range and candidate-generation behavior may change according to equipment position and the type of inheritance source.

5-1. Nested Structures and Definitions of Recursive Processing

ONE-LINE SUMMARY Defines recursive reference to arrangement information held inside inherited materials and identifies the questions that require validation.

In the Rune Factory series, Performance Inheritance, Arrangement Inheritance, and Auto Arrange interact in complex ways. A particularly important question is what happens when equipment that already inherited an arrangement is used again as an inheritance material.

In this document, “Recursive Processing” and “Nested Reference” are convenient observation labels for processing in which arrangement information, Auto Arrange information, or internal material information retained inside an inheritance material is referenced again or becomes a candidate.

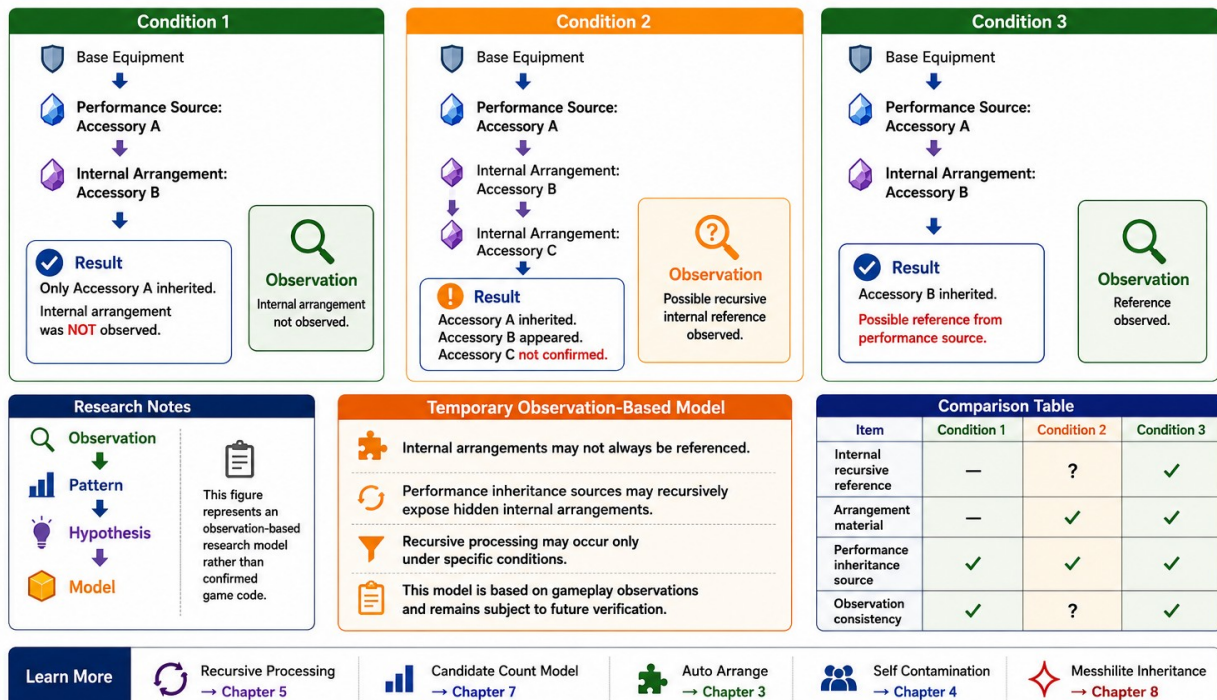
For a nested structure such as Equipment A → Equipment B inside A → Equipment C inside B, the questions are whether only A is referenced, whether B also becomes a candidate, and whether C is recursively expanded.

The internal algorithm is unresolved. This chapter is a provisional, observation-based organization and does not establish an implementation.

RF-MAP-05

RF5 Recursive Processing Concept

One-line Summary: Performance inheritance may recursively reference internal arrangement data.
(This is an observation-based research model.)



Observation-oriented figure. Scope and confidence follow the surrounding section.

Target element	Recursive-reference possibility	Current observed tendency	Section
Performance-source equipment	High possibility	Internal arrangements may become candidates or expand	5-2
Additional arrangement material	Low possibility	Internal Auto Arrange expansion not observed	5-3
Internal Auto Arrange material	Possible	May affect the candidate group	3 / 5-6

05 Figure Identity / Numbering Record

Document identity patch; semantic authority is evaluated separately.

Figure 5-1	Recursive-reference scope
Figure 5-2	Performance-Source-side observation
Figure 5-3	Additional-arrangement-side comparison
Figure 5-4	Internal-material candidate model
Figure 5-5	Recursive-depth question
Figure 5-6	Retained Auto Arrange relationship
Figure 5-7	Practical stage separation

		depending on conditions	
Performance-Source Contamination candidate	Possible	May re-enter as a candidate originating from the inheritance source	4
Inherited arrangement information	High possibility	May be referenced again during later inheritance	5-2
Base-equipment appearance	Appears low	Limited effect in observations so far	2
Candidate-order information	Unknown	Order variability and candidate-order changes observed	4-6
Deep nested material	Unresolved	Maximum recursive depth unconfirmed	5-5

Boundary note: provisional observation-based organization; not an internal-algorithm claim and not guaranteed across all conditions.

5-2. Reference from the Performance-Source Position

ONE-LINE SUMMARY Observations suggest that internal Auto Arrange materials in Performance-Source equipment have a relatively high possibility of becoming candidates.

RF5 tests used equipment with internal Auto Arrange information, including the Hand-Knit Scarf and Ghost Boots. Observations were compatible with internal arrangement information being referenced when that equipment occupied the Performance Source position.

Example condition: base equipment = Field Gloves; Performance Source = arranged Hand-Knit Scarf; additional arrangement = Talisman. Patterns were observed in which the Talisman and internal materials 1 and 2 became candidates or were expanded.

Other observed outcomes included the Talisman remaining, the Talisman dropping out, and the performance of the Hand-Knit Scarf itself becoming a candidate.

Within the tested scope, information on the Performance-Source side may receive a special reference treatment. This is a possibility indicated by observation, not a universal rule.

RF5 Observation Model

Internal Arrangement Reference Hypothesis

Observation-Based Explanatory Model (Hypothesis)

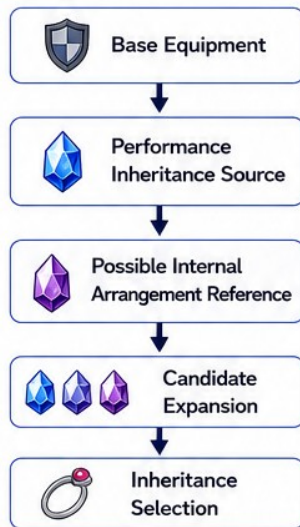


Core Concept

- The Performance Inheritance Source may reference its own internal arrangement.
- Internal arrangements may recursively expand the effective candidate pool.
- Additional arrangement materials appear less likely to be recursively referenced.
- The actual internal game algorithm has **NOT** been confirmed.

Everything shown is an observation-based explanatory model, not verified game code.

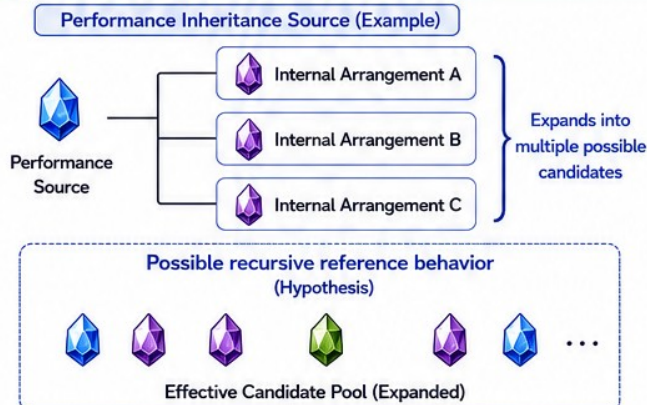
1 Overall Flow



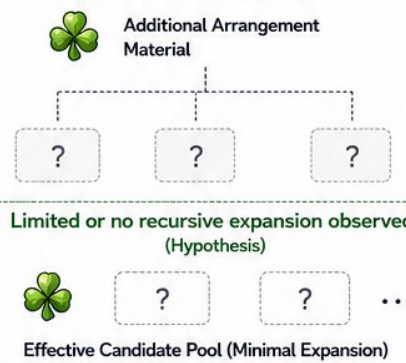
2 Observation Summary

	Performance Inheritance Source	Additional Arrangement Material
Internal arrangement reference possibility	✓ May be referenced (Observed possibility)	? Not confirmed / Appears less likely
Recursive expansion of candidates	✓ Observed in cases	? Limited evidence
Consistency across observations	✓ Multiple observations support the model	? Inconsistent / Insufficient data
Current confidence level	★★★☆☆ (Moderate)	★☆☆☆☆ (Low)

3 Illustrative Example (Hypothesis)



Additional Arrangement Material (Example)



4 Current Interpretation



- ✓ Performance inheritance sources may recursively reference internal arrangements.
- ✓ Additional arrangement materials currently show limited evidence of recursive reference.
- ✓ Candidate expansion may depend on equipment position.
- ✓ Further verification is required.



Observation-Based Research

- This figure summarizes gameplay observations and explanatory models.
- The internal game algorithm has not been confirmed.
- Interpret this figure as an observation-based hypothesis rather than verified game code.

Observation-oriented figure. Scope and confidence follow the surrounding section.

5-3. Tests on the Additional-Arrangement-Material Side

ONE-LINE SUMMARY Internal Auto Arrange information in an additionally inserted material appears less likely to be recursively expanded.

RF5 condition: base equipment = Field Gloves; Performance Source = Talisman; additional arrangement = Hand-Knit Scarf with internal Auto Arrange information. The Talisman and Hand-Knit Scarf were observed, but the internal Auto Arrange materials of the scarf were not observed.

RF4SP condition: base equipment = Piyo Sandals; Performance Source = Leather Boots; additional arrangement material = Feather Boots. Leather Boots and Feather Boots were observed, but the internal Auto Arrange materials of Feather Boots were not observed.

Within these observations, the additional-arrangement-material position appears less likely to recursively expand internal information than the Performance-Source position. This result remains condition-bound.

Additional Arrangement Material: Observed Boundary

RF5 observation

The directly inserted item was observed, but its internal Auto Arrange materials were not observed. This suggests that recursive expansion may depend on placement.

RF4SP observation

Leather Boots and Feather Boots were observed. Internal Auto Arrange materials of Feather Boots were not observed in the tested condition.

Observation boundary: not a proof of universal non-expansion.

Observation-oriented figure. Scope and confidence follow the surrounding section.

5-4. Internal-Material Candidate Model

ONE-LINE SUMMARY Internal candidate generation may depend on position, with the Performance Source as the main reference position.

The observations fit a position-dependent model better than a model in which every item is uniformly expanded. Current provisional organization: the Performance Source may be referenced; recursive reference from an additional arrangement material is unconfirmed and appears less likely.

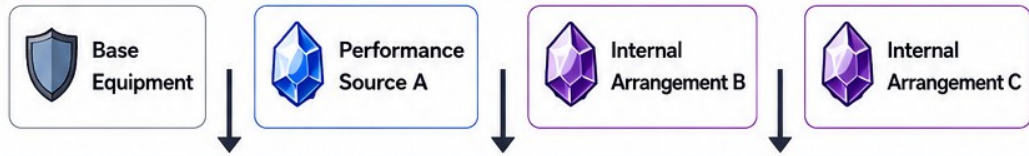
This candidate-generation behavior may connect to Performance-Source Contamination, candidate-count growth, order variability, and Auto Arrange. Candidate-generation timing, reference priority, and internal draw order remain unresolved.

Candidate Expansion Example

Observation-Based Candidate Pool Expansion Model

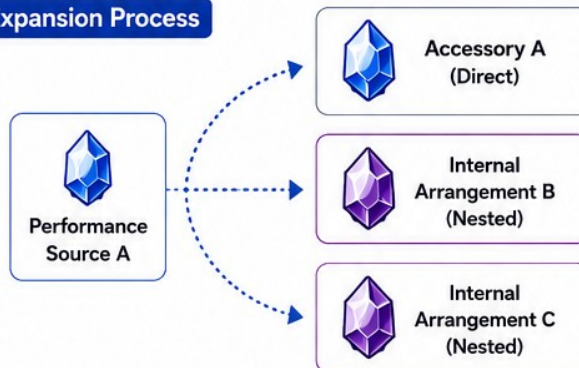
1 Original Equipment Structure

Original inheritance structure



2 Candidate Expansion Process

Performance Source A expands into multiple potential candidates.



Candidate Expansion (Hypothesis)

Nested internal arrangements may be referenced as additional candidates.

This is an observation-based hypothesis, not confirmed internal code behavior.

3 Observed Candidate Pool

Observed candidate pool after expansion

Possible recursive reference behavior



Direct candidates

Candidates from nested internal arrangements

4 Observation-Based Interpretation



- ✓ Internal arrangements may expand into additional candidates.
- ✓ Nested arrangements may increase the effective candidate pool.
- ✓ Expansion behavior appears observation-dependent.
- ✓ Additional verification is required.

Key Insight

One inheritance source may recursively expand into multiple effective candidates.

Candidate Expansion Characteristics

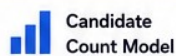
Item	Observation
Direct inheritance	✓
Nested reference	Observed
Recursive expansion	? Possible
Confirmed by code	✗ No



Observation-Based Research Model

- This figure summarizes gameplay observations and explanatory models.
- The actual internal algorithm has not been confirmed.
- Interpret this figure as an observation-based hypothesis rather than verified game code.

Related Concepts



Candidate Count Model



Recursive Processing



Self Contamination



Performance Inheritance

Icon Legend



Blue Gem
= Performance Source



Purple Gem
= Internal Arrangement



Green Gem
= Additional Arrangement



Gray Shield
= Base Equipment

Observation-oriented figure. Scope and confidence follow the surrounding section.

Position	Internal-information reference	Possible candidate effect	Observed tendency	Note
Performance Source	Relatively high possibility	Internal arrangement materials may be added to the candidate group	Observed in RF5 Condition 2	Primary position-dependent candidate
Additional arrangement material	May be less likely	Internal candidates unconfirmed	Not observed in RF5 Condition 1 or RF4SP Feather Boots test	May be treated only as a direct input
Base-equipment side	May be relatively unlikely	Likely small candidate effect	Limited effect so far	Separate appearance from performance
Internal Auto Arrange information	Condition-dependent	May affect candidates when held by the Performance Source	Suggested by RF5 Condition 2	Connects to Ch. 3 and 5-6
Inherited arrangement information	Possible	May be registered again in later inheritance	Connects to Ch. 4 observations	Related to contamination and candidate growth
Deep internal material	Unresolved	Whether it becomes a candidate is unconfirmed	Insufficient direct testing	Connects to recursive depth

Boundary note: provisional observation-based organization; not an internal-algorithm claim and not guaranteed across all conditions.

5-5. The Recursive-Depth Question

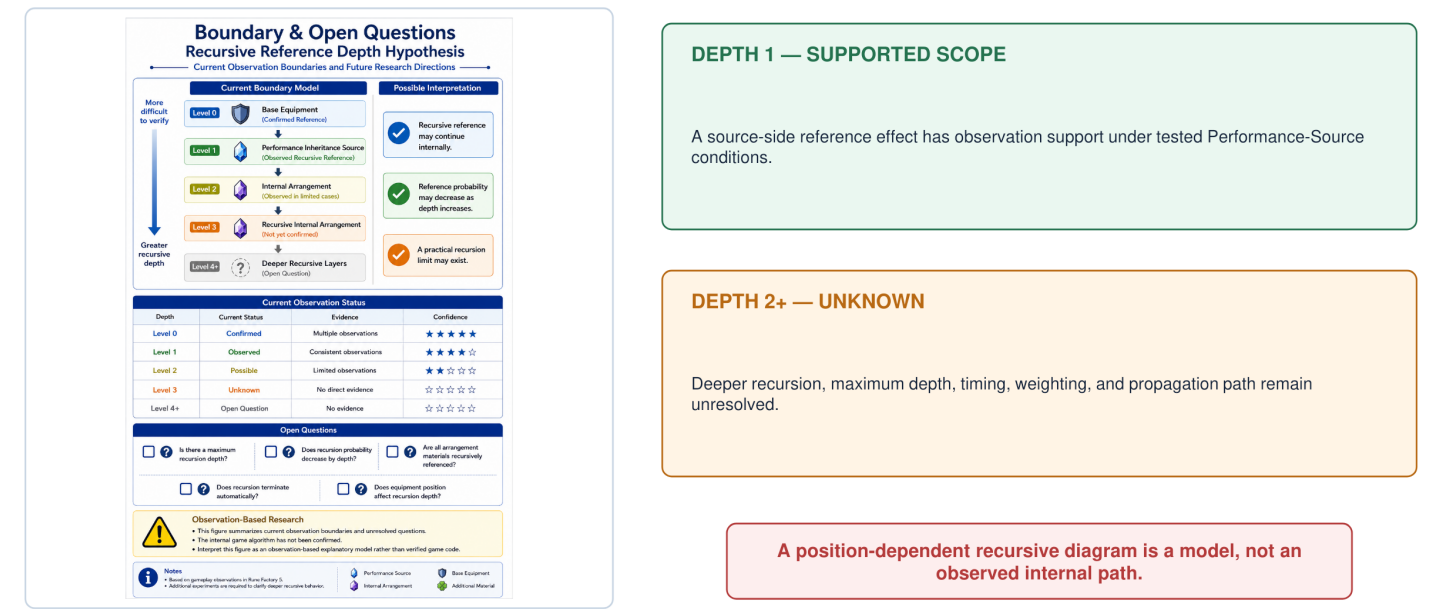
ONE-LINE SUMMARY Whether a recursive-depth limit exists is unconfirmed; depth 2 and beyond require direct testing.

How deeply internal information can be referenced remains unresolved. In a structure $A \rightarrow B \rightarrow C$, it is unknown whether B is referenced, whether C is expanded, or whether a depth limit exists.

Important future tests include structures with three or more layers, accessory inheritance, and combinations with Messhilite inheritance.

Figure 5-4

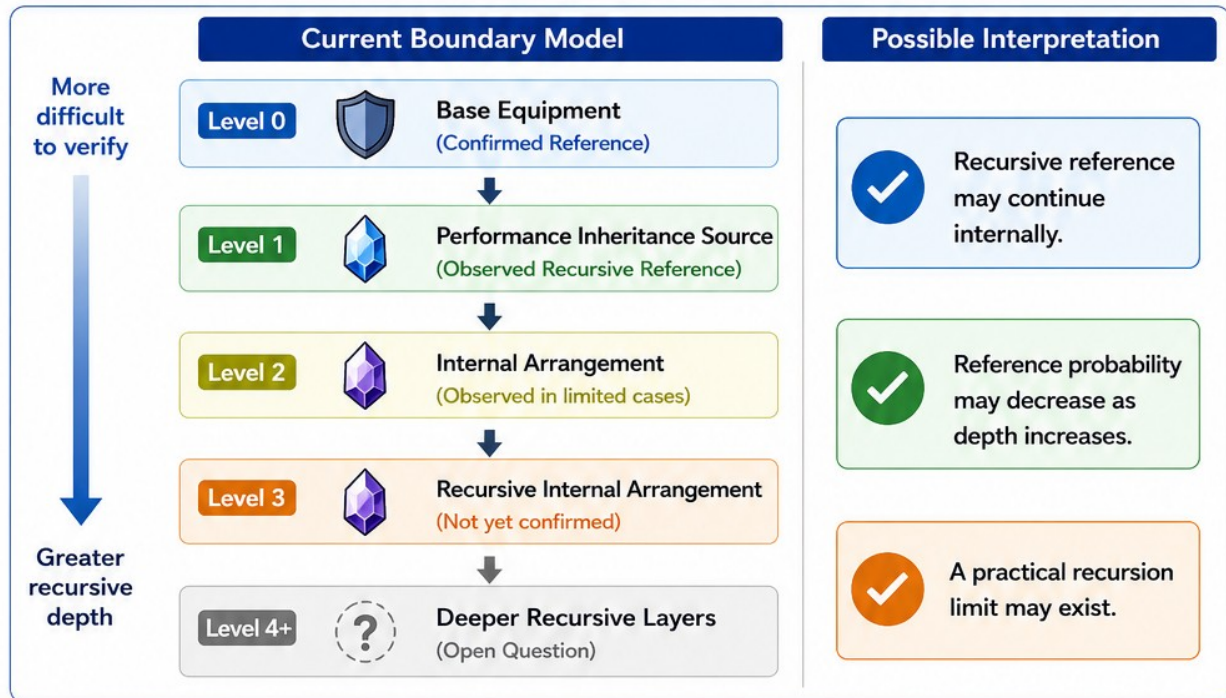
Recursive Processing: Depth-1 Support Does Not Establish Deeper Recursion



Boundary & Open Questions

Recursive Reference Depth Hypothesis

Current Observation Boundaries and Future Research Directions



Current Observation Status			
Depth	Current Status	Evidence	Confidence
Level 0	Confirmed	Multiple observations	★★★★★
Level 1	Observed	Consistent observations	★★★★☆
Level 2	Possible	Limited observations	★★★☆☆
Level 3	Unknown	No direct evidence	☆☆☆☆☆
Level 4+	Open Question	No evidence	☆☆☆☆☆

Open Questions

☐ ? Is there a maximum recursion depth?

☐ ? Does recursion probability decrease by depth?

☐ ? Are all arrangement materials recursively referenced?

☐ ? Does recursion terminate automatically?

☐ ? Does equipment position affect recursion depth?



Observation-Based Research

- This figure summarizes current observation boundaries and unresolved questions.
- The internal game algorithm has not been confirmed.
- Interpret this figure as an observation-based explanatory model rather than verified game code.

 Notes

- Based on gameplay observations in Rune Factory 5.
- Additional experiments are required to clarify deeper recursive behavior.

 Performance Source

 Internal Arrangement

 Base Equipment

 Additional Material

Observation-oriented figure. Scope and confidence follow the surrounding section.

Depth	Example	Observation status	Representative observation	Current organization
0	Equipment A	Confirmed	Ordinary inheritance	Consistently becomes a candidate
1	$A \rightarrow B$	Confirmed	RF5 Condition 2	May be referenced again from the Performance-Source side
2	$A \rightarrow B \rightarrow C$	Unconfirmed	No direct test	May become a candidate depending on conditions
3	$A \rightarrow B \rightarrow C \rightarrow D$	Unconfirmed	No direct test	A depth limit may exist
4+	Multiple nested layers	Unconfirmed	Untested	Maximum depth unknown

Boundary note: provisional observation-based organization; not an internal-algorithm claim and not guaranteed across all conditions.

5-6. Relationship to Retained Auto Arrange Information

ONE-LINE SUMMARY Information generated through Auto Arrange may affect later inheritance depending on conditions.

Recursive Processing may be closely related to information generated by Auto Arrange, inherited arrangement information, and candidates originating from Performance-Source Contamination.

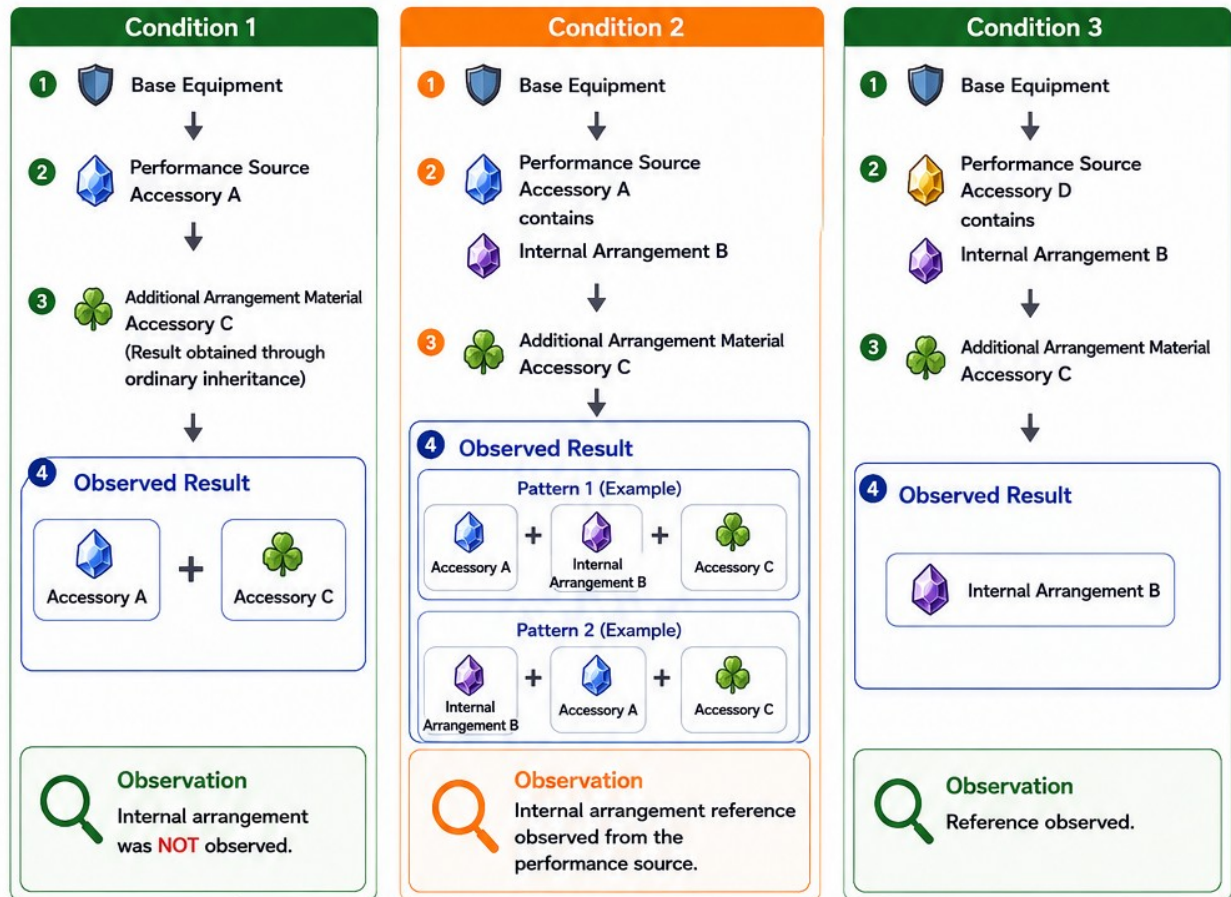
Within the observed scope, an internal arrangement created once may affect later inheritance depending on conditions. It is unlikely that all internal information is expanded without limit; equipment position, inheritance type, and candidate-generation conditions may control the reference range.

Recursive Reference Observation

Observation-Based Internal Arrangement Reference Comparison



Gameplay observations suggest that internal arrangement data may be referenced from the performance inheritance source under specific conditions.



Observation-Based Interpretation



- Internal arrangements inside additional arrangement materials were not observed.
- Internal arrangements inside performance inheritance sources may be referenced.
- Recursive reference behavior may depend on inheritance source structure.
- Additional observations are required.

Comparison Summary

Item	Additional Arrangement Material	Performance Inheritance Source	Observation Consistency
Internal arrangement reference	—	✓	?
Reference of nested arrangement data	—	✓	?
Recursive reference behavior	—	✓	?
Observation consistency	✓	?	?

Observation Notes

- ✓ Internal reference has been observed.
- ✓ Reference behavior differs by source type.
- ✓ Recursive reference may occur under specific conditions.
- ✓ This figure summarizes gameplay observations only.

Related Concepts



Recursive Processing



Candidate Count Model



Self Contamination



Auto Arrange



Observation-Based Research Model

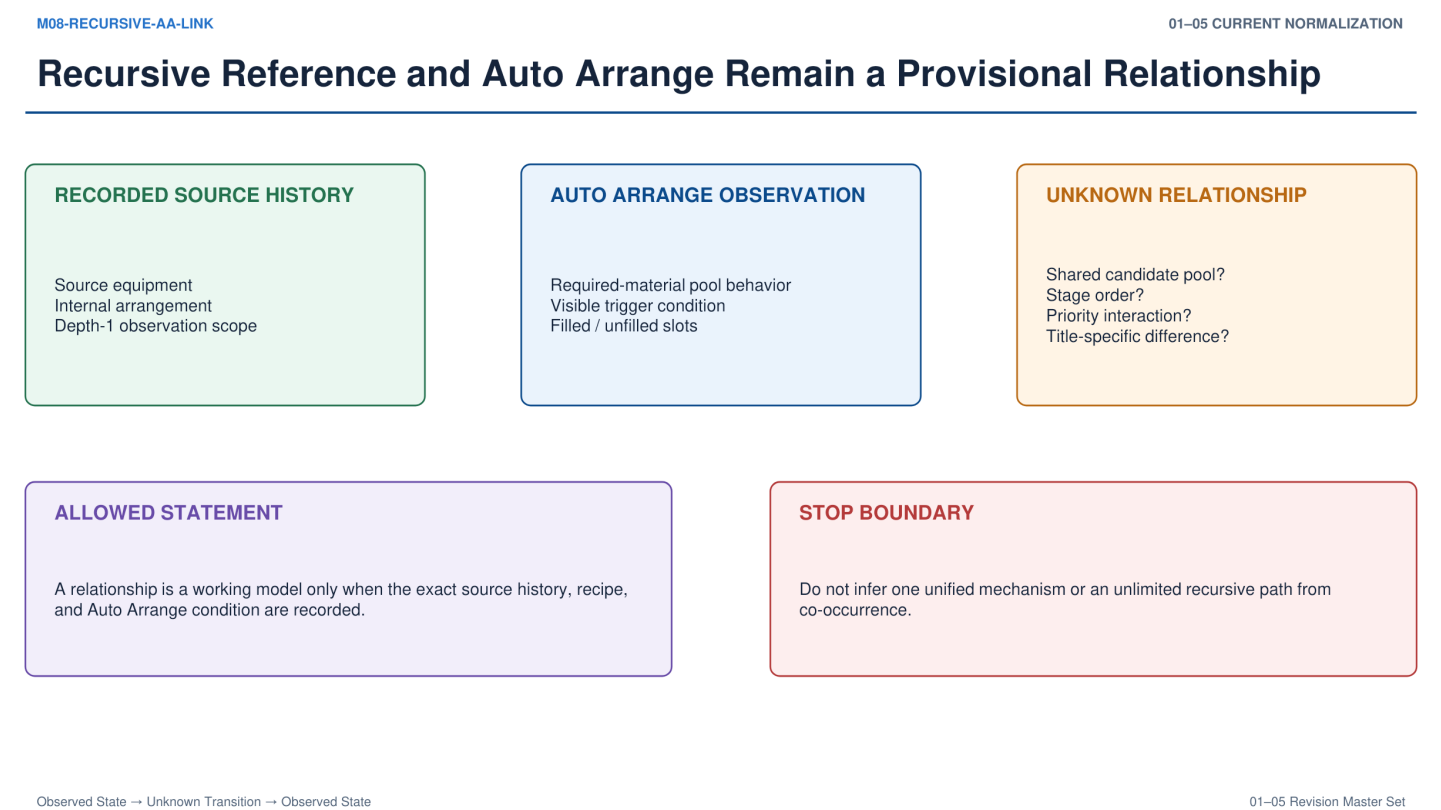
- This figure summarizes gameplay observations and explanatory models.
- Observed recursive reference behavior may depend on equipment structure and inheritance conditions.
- The actual internal game algorithm has not been confirmed and remains under investigation.
- Avoid interpreting this figure as confirmed game code.

Icon Legend

- Accessory A / D
- Internal Arrangement B
- Additional Arrangement C
- Base Equipment

Observation-oriented figure. Scope and confidence follow the surrounding section.

Figure 5-6



Observation item	RF4SP	RF5	Current observed tendency	Related section
Auto Arrange occurrence	Observed	Observed	Occurs under required-material and open-slot conditions	3
Auto Arrange slot increase	Observed	Observed	Affects candidate-count growth	3 / 4
Order variability	Observed	Observed	Final display order may not be fixed	4-6
Relationship to contamination	Possible	Possible	Retained information may become a candidate again	4
Internal reference from Performance Source	Unconfirmed	Observed	Special reference may occur only on that side	5-2
Internal expansion from additional material	Unconfirmed	Unconfirmed	Internal materials may be less likely to expand	5-3
Internal-material candidate generation	Possible	Observed	Retained internal information may join the candidate group	5-4
Recursive-depth limit	Unresolved	Unresolved	A maximum depth may exist	5-5
Candidate-order variation	Observed	Partially observed	Possible shuffle after selection	4-6 / 6
Persistence of Auto Arrange information	Possible	Possible	May affect later inheritance	5-6
Compatibility with split saves	Potentially useful	Potentially useful	May help fix an earlier stage	8

Boundary note: provisional observation-based organization; not an internal-algorithm claim and not guaranteed across all conditions.

5-7. Effects on Split Saves and Stage Separation

ONE-LINE SUMMARY Controlling recursive candidate generation may make split saves and separated crafting stages useful in practical high-difficulty inheritance.

If this provisional organization is correct, an arrangement completed in an earlier stage may be treated as a comparatively stable asset. This affects split saves, stage separation, high-difficulty inheritance, and Messhilitite inheritance.

If internal information on the additional-arrangement-material side does not collapse or expand without limit, a candidate group built in a previous stage may remain somewhat stable.

However, the number of trials is insufficient, boundary values are untested, and not all equipment types have been checked. The following is a provisional practical organization, not an optimum or an internal-specification claim.

I was able to turn one RNG step into a guaranteed outcome in this crafting route. By using an item with no internal arrangement as the first material, some routes can reduce the number of required resets. I call this approach the “Cushion Method.” Based on gameplay observations.

Quiz! These two crafting setups use the same materials. Which one do you think requires RNG? Answer: The one on the left. The order of the materials can affect whether a reset is required. I call this approach the “Shuffle Method.” It is an application of the Recursive Processing observations.

Quiz #2. Which of these two crafting setups do you think requires RNG? The answer is the one on the right. The setup on the left can actually avoid the RNG. Can you spot the difference? The left accessory was obtained from Terry’s return gifts, while the right one was player-crafted. Player-crafted equipment may contain internal arrangements created by Auto Arrange. As a result, the candidate pool can exceed three, making RNG necessary. This is one example of how hidden internal arrangements can affect inheritance results. Based on gameplay observations. This does not claim to prove the game’s internal implementation.

Shuffle Method and Cushion Method

Shuffle Method

Even with an apparently identical material set, changing the first material may change internal references, candidate count, and which step requires resetting.

Cushion Method

Place equipment with no internal arrangement first. In some routes, this can turn one RNG-dependent checkpoint into a deterministic step and reduce reset conditions.

Practical method based on in-game observation; not an internal implementation claim.

Observation-oriented figure. Scope and confidence follow the surrounding section.

Element	Practical effect	Potential benefit	Potential risk / issue
Split save	Fix an earlier stage	Reduces rebuilding burden	More stages and management
Intermediate equipment	Organize candidate group	May stabilize construction step by step	May add internal candidates
Recursive Processing	Internal candidate generation	May preserve special inheritance structures	Candidate growth and lower success rate
Auto Arrange	Fill missing slots	Inheritance may succeed with insufficient direct materials	Unexpected candidates may enter
Performance-Source Contamination	Candidate re-registration	May enable special structures	Lower success and more order variability
Order variability	Candidate-order changes	Avoids dependence on fixed order	Harder to target
Candidate-count growth	Harder draw	More complex combinations become possible	Success rate may fall sharply
Stage separation	Distribute work	Manage high-cost inheritance in stages	Complex save management
Messhilita inheritance	Extremely difficult structure	Enables special inheritance	Candidate explosion and worse expectation
Role-play equipment	Preserve appearance	Combine world-building, appearance, and performance	Higher trial burden

Boundary note: provisional observation-based organization; not an internal-algorithm claim and not guaranteed across all conditions.

Source and Addition Boundary

This full translation preserves the Japanese Canonical chapter's observation strength and unresolved boundaries. The Shuffle Method and Cushion Method passages are user-authorized post-Canonical additions supplied on 2026-08-08. They are practical observations, not retroactive claims about the original chapter or the game's internal implementation.